Kilcarrow Works - memo

To: Executive team. From: Naomi Lynskey, Environment and Quality. Date: 11 January 2027.

Subject: mercury to air in 2026 against the 85 kg annual load from 2028 (licence condition 5.6)

At its 2026 release the works would not keep under the 85 kg load. Kiln 1 released 68 kg of mercury to air in 2026 and kiln 2 released 83 kg, 151 kg for the works, 66 kg over the load. Carbon injection, or another measure of the same size, is needed before 2028; it belongs in the 2027 capital round and in the reduction programme due to the Agency by 31 March.

How the figure was set

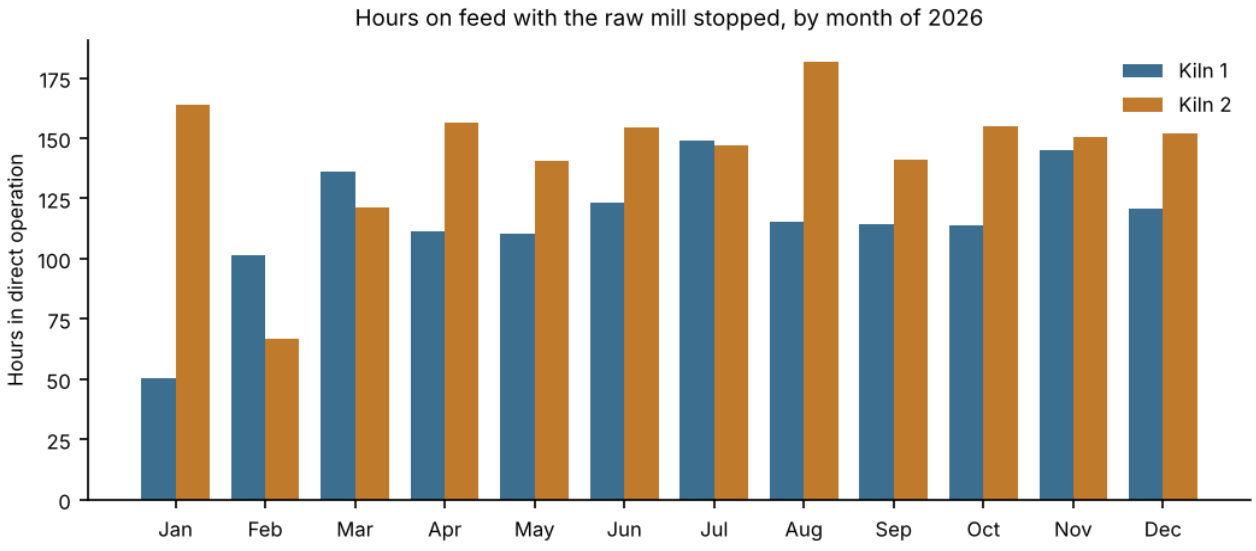
Mercury leaves a kiln line three ways: in clinker, in filter dust taken out of the system, and through the stack. Over a year the mercury the line took in with its raw materials and fuels is what left by those routes, so the release to air is the year's input less what went out in clinker and in dust. Inputs are the production ledger's monthly tonnages (raw materials dry, fuels as fired) times the laboratory's mercury composite for the month on the same basis (sand and slag: the quarter's composite); clinker is the ledger's clinker times its composite; kiln 1's dust is the ledger's filter dust to silo S7 times the month's dust composite. Kiln 2 returns its dust to the kiln feed.

	Kiln 1	Kiln 2
Mercury in raw materials and fuels, kg	80.8	84.1
Less mercury in clinker, kg	1.3	0.9
Less mercury in filter dust bled to silo S7, kg	11.6	-
Released to air, kg	**68**	**83**

Sources: production ledger 2026 (raw materials dry, fuels as fired); QC laboratory composites 2026.

Why the stack tests give less

Every annual mercury test (2024 to 2026) ran with the raw mill on, when the mill takes the kiln gas's mercury into the raw meal and hands it back to the kiln. The tests agree with each other, but they measure that state only: carried over the year's kiln hours they give about 70 kg for the works. Condition 5.6 counts every hour, and with the mill stopped (direct operation) the gas goes straight to the filter and the stored mercury goes with it. The chart shows each kiln's hours in direct operation by month.



Other 2026 figures for the annual report (NOx, hours on feed and hours in direct operation by quarter) are in kiln_emissions_2026.xlsx.

Note: direct operation is as defined in licence condition 1.5 (kiln producing clinker, raw mill stopped); hours are counted from the control system journal in Irish local time.